TECHNICAL ARCHITECTURE

Emouva

AI-powered investment intelligence that tells you what your broker won't — what's actually wrong with your portfolio and what to do about it.

React + TypeScript | FastAPI + Python | Claude AI (Opus 4.6) | PostgreSQL | Robinhood OAuth + MCP | Agentic Trading

LIVE — EMOUVA.COM

THE PROBLEM

600M+ retail investors make decisions without intelligence

Data, Not Judgment

Every competitor gives you charts, numbers, and metrics. None tells you **what it means for your specific portfolio**.

No Advisory Loop

"Your portfolio is risky" — ok, **but what do I do about it?** No consumer app closes the loop from diagnosis to action.

Hidden Complexity

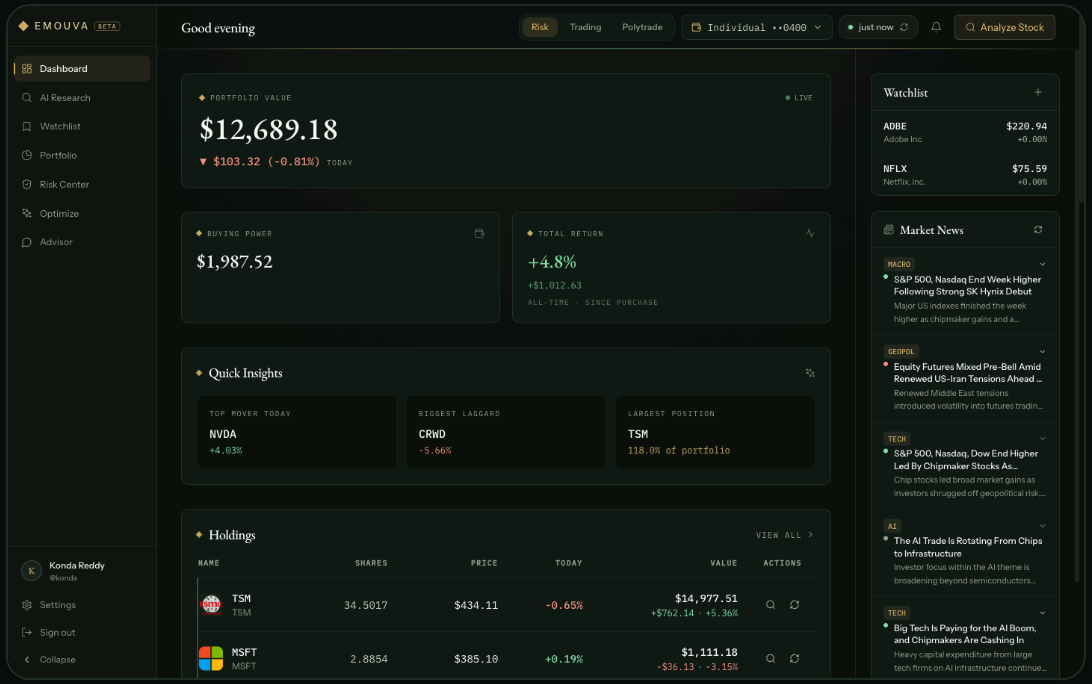
Correlation risk, sector concentration, stress scenarios — **concepts a $5k investor deserves** but can't access without a $10k/yr advisor.

Emouva gives judgment, not data. In plain language, personalized to your actual portfolio.

Full advisory loop: diagnose risk → show what could go wrong → recommend what to do → execute if you want.

Live portfolio intelligence in one view

- **Real-time positions** with sparklines, P&L, and sector data from Robinhood
- **Quick Insights** — top mover, biggest laggard, largest position at a glance
- **AI-powered columns** — Fair Value, Rating, Sentiment per stock (cached, not re-computed)
- **Watchlist + News Feed** with urgency badges and ticker tags
- **Buying Power**, Total Return, Risk Score summary cards



SYSTEM ARCHITECTURE

Three-layer architecture with shared intelligence

Frontend
React + TypeScript + Vite

→

FastAPI Backend
Python 3.11+ • Async

→

External Services
APIs + AI

PAGES

Dashboard • AI Research • Portfolio
Risk Center • Diversify • Stress Test
Advisor • Trading Suite • Polytrade

CREDENTIAL STORE

BYOK — API keys stay in browser
localStorage (base64, never server-stored)

SERVICES

Claude AI • Risk Engine • Market Data
Stock Cache • Robinhood OAuth/MCP
Agent Harness • Theme Engine

DATA

PostgreSQL (44 models)
Per-field JSONB cache + TTLs
Append-only audit journal

AI

Claude Opus 4.6 (research, theses)
Claude Sonnet 4.6 (sentiment, advisor)
Web search tool integration

MARKET DATA

Robinhood (positions, quotes, orders)
yfinance • FMP • Finnhub (fallback chain)

PATTERN

Stale-While-Revalidate — serve cached data instantly, refresh in background. Progressive SSE loading for cache misses.

SECURITY

BYOK model — user API keys never stored server-side. Robinhood OAuth tokens encrypted at rest. Agentic account fenced.

SCALE

44 SQLAlchemy models, 16+ API routers, per-field JSONB cache with TTLs from 1 min (price) to 7 days (AI analysis).

AI STOCK RESEARCH PIPELINE

From ticker to investment thesis in seconds

1. **Enrichment** — Pull 30+ metrics from yfinance/FMP: P/E, beta, earnings, sector, fundamentals. Buffer in memory.
2. **Claude Analysis** — Opus 4.6 with web_search tool. Generates DCF valuation range (bear/fair/bull), investment thesis, and risk factors.
3. **Bear Case Stress Test** — Adversarial red-team of the bull thesis. Custom scenario input (e.g., "What if their CEO resigns?").
4. **Sentiment Score** — 4 dimensions (News, Filings, Insider, Analyst) composited to a score out of 100.
5. **Write-through Cache** — Results persisted to `stock_metrics` (JSONB) via BackgroundTasks. 7-day TTL for AI, 24h for sentiment.

SHARED INTELLIGENCE CACHE

METRIC	TTL	SOURCE
Price	1 min	Robinhood
Company Info	24 hours	yfinance / FMP
News & Sentiment	24 hours	Claude AI
Earnings	90 days	yfinance
Fair Value / Thesis	7 days	Claude Opus
Bear Case	7 days	Claude Opus

SWR Pattern: Any user searching a ticker gets instant results from pre-computed data. Stale metrics refresh in background — zero wait for popular tickers.

RISK ANALYTICS ENGINE

Institutional-grade risk analysis for every investor

61	-2.3%	34%	-18%
RISK SCORE /100	DAILY VAR (95%)	ANNUALIZED VOL	MAX DRAWDOWN
Moderate-High	Historical simulation	$\sigma \times \sqrt{252}$	Peak-to-trough

Composite Risk Score Formula

```
risk_score =
  VaR_score × 0.25
+ Drawdown_score × 0.20
+ Concentration_score × 0.20
+ Volatility_score × 0.20
+ Correlation_score × 0.15
```

Concentration Risk (HHI)

Three dimensions: **Sector** (45%), **Market Cap** (30%), **Geography** (25%)

HHI = $\Sigma(\text{weight}_i)^2$ — Low (<0.15) / Moderate / High (>0.25)

30 Stress Test Scenarios

Historical
2008, COVID, Dot-com, Black Monday

Macro
Rate Hike, Recession, Stagflation

Geopolitical
China-Taiwan, Trade War, Oil Shock

Sector
AI Bubble, EV Bubble, Bank Run

Per-Stock Impact Model

```
impact = base_sector_impact
  × beta_multiplier
  × size_multiplier
  × quality_multiplier
  + correlation_spike_adj
```

Cold-start AI agent with hard safety gates

Decision Loop (per tick)

1. Wake — Scheduler fires (5m / 15m / 30m / 1h / Daily)

↓

2. Load State JIT — Policy + Strategy + Portfolio Snapshot + Last 5 Decisions

↓

3. Brain.decide() — Cached prefix (creed + tools) + small dynamic tail (~300 tokens)

↓

4. SafetyGate — 9 hard code-level gates (NOT prompted)

↓

5. Execute — Paper (synthetic) | Alpaca (paper API) | Live (Robinhood MCP)

↓

6. Journal + Memory — Append-only JSONL audit, memory files for next tick

Context Efficiency

Each tick = **one fresh query** to Claude (NOT a resumed session). Context window is scratch space, not storage.

~482
TOKENS / TICK

0.1x
COST ON CACHE HIT

Cached Prefix (1h TTL)

Agent Creed (Munger principles) + **System Policy** + **Decision Tool Schema** → paid ~0.1x on repeat ticks.

Dynamic Tail (bounded)

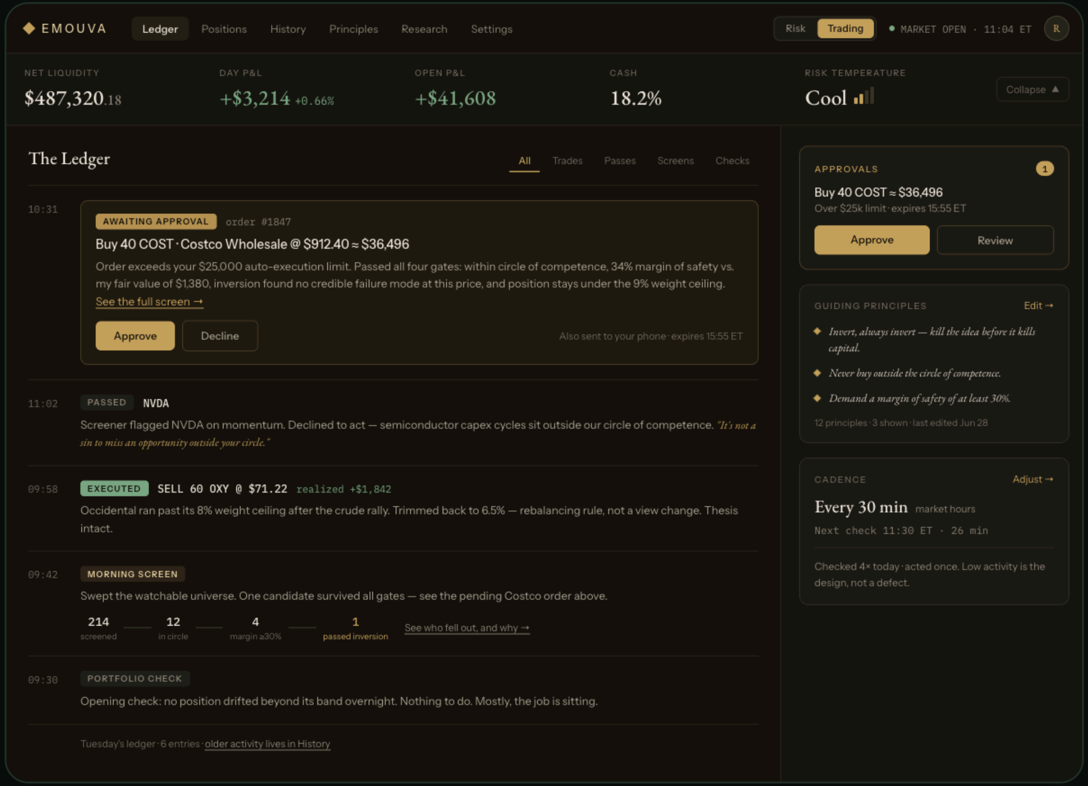
Portfolio snapshot + 5-decision recap + active theses + live quotes. **Flat input size** regardless of how many ticks have run.

SAFETY & GUARDRAILS

9 hard gates in code — not in the prompt

1. **Kill Switch** — User can disable agent instantly. Fail-closed.
2. **Allowlist** — Only trade symbols on the user's approved list.
3. **Per-Trade Cap** — Max $250/order (configurable by user).
4. **Daily Spend Cap** — Total daily buy spend limit.
5. **Position Concentration** — No single stock > 10% of equity.
6. **Funds Check** — Cannot exceed buying power.
7. **Cash Floor** — Always keep 10% in cash.
8. **Sector Cap** — Max 25% allocation to any sector.
9. **Trading Pace** — Max 3 orders/week. Prevents overtrading.

Approval Threshold: Orders above $250 require explicit user approval via push notification + approval card. Agent pauses until confirmed.



The Ledger — Full Audit Trail

- **Approve / Decline** inline on pending orders
- **Morning Screen** — universe sweep with gate results
- **Every decision** logged: proposal, gate result, execution, cost
- **Immutable JSONL** — append-only journal per user

Pre-reasoned opportunities shared across all agents

Opportunity Pool

Admin-curated + AI-researched stock analyses. Each entry contains: **fair value estimate**, **investment thesis**, **composite score**, and **category** (tradeable / hard to understand / rejected).

One symbol → one analysis. Reused by all users' agents — not re-derived per account.

Screening Flow

Admin seeds candidates (ticker + rough narrative)

↓

Claude researches: DCF, thesis, falsifiers, score

↓

Category 1 (tradeable) enters agent universe

↓

Agent applies policy + safety gates at trade time

Living Theses

Every position has a **monitorable thesis**: narrative + machine-checkable falsifiers (measurable metric + threshold). The agent evaluates thesis health each tick.

```
ACTIVE → thesis holds, stay long
FLASHED → falsifier weakened, watching
BROKEN → exit signal, sell queued
```

Stock Metrics Cache

Shared intelligence layer: per-ticker, per-field JSONB cache with individual TTLs. Used by Portfolio table (Fair Value, Rating, Sentiment columns), Dashboard, and the agent's research view.

`stock_metrics → market_data (15m) | company_info (24h) | earnings (90d) | ai_analysis (7d)`

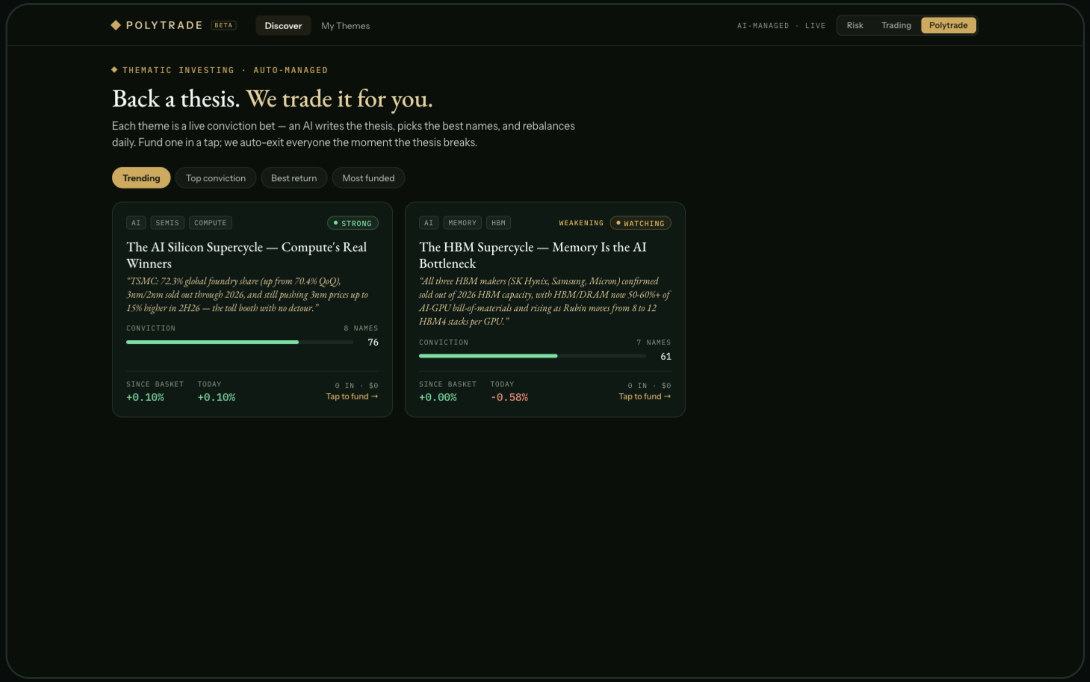
POLYTRADE — THEMATIC BASKETS

Back a thesis. We trade it for you.

AI-curated, thesis-driven baskets. Users fund a theme in one tap — Emouva auto-invests, monitors daily, rebalances on drift, and **auto-exits everyone the moment the thesis breaks**.

Five Pipelines

- A **Thesis Origination** — Admin seeds narrative, Claude researches with web_search, drafts falsifiers + red-team.
- B **Basket Construction** — 4-8 stocks picked (anchor / satellite / speculative roles). Conviction-weighted, guardrails enforced.
- C **Daily Monitoring** — Falsifier check at 7am, material news flags intraday, deep re-validation weekly.
- D **Earnings Triggers** — Constituent reports → theme-scoped re-thesis. Update conviction or trip falsifier.
- E **Propagation** — Version-driven reconciler walks every allocation. Idempotent: `(allocation_id, target_version)`.



BASKET RULES

Per-name ≤ 35%
Speculative sleeve ≤ 20%
Reweight on > ±5pp drift

ISOLATION

Theme holdings live in separate book from discretionary positions

Closing the advisory loop — diagnosis to action

Behavioral Risk Score

COMPOSITION (40%)
53 / 100

CONCENTRATION (30%)
49 / 100

VOLATILITY (20%)
34 / 100

CORRELATION (10%)
100 / 100

Key Findings (plain English)

- "82% of your money is in tech. That's a big bet on one sector."
- "Portfolio beta of 1.38x the market. When S&P drops 10%, you drop ~14%."
- "20 highly correlated pairs. Your stocks move together."
- "Good sector spread — 5 sectors represented."

Before vs After Simulation

PROTECTED IN A 2022-STYLE CRASH

$807

with 3 simple changes

SCENARIO	BEFORE	AFTER
2022 Rate Shock	-$2,841	**-$2,034**
COVID 2020	-$1,892	**-$1,461**
Health Score	48	**78**
Sectors	5	**8**

Suggested Additions

XLE	Energy — Best crash savings +$544
XLV	Healthcare — Best crash savings +$166
BND	Bonds — Best crash savings +$291

44 models, 16 routers, 3 execution modes

Stack

Frontend
React 18 + TypeScript + Vite
Tailwind CSS + Recharts

Backend
FastAPI + SQLAlchemy 2.0
Python 3.11+ Async

AI
Claude Opus/Sonnet 4.6
Web Search + Prompt Cache

Broker
Robinhood OAuth 2.1 + PKCE
Agentic MCP

Core Data Model (PostgreSQL)

DOMAIN	TABLES	PURPOSE
Auth	users, robinhood_connections	Accounts + OAuth tokens (encrypted)
Portfolio	accounts, account_positions, deposits	Multi-account (paper/agentic/RH)
Intel	opportunities, theses, stock_metrics	Central pool + living theses + cache
Agent	mandates, orders, ticks, ledger	Config + decisions + audit trail
Polytrade	themes, constituents, allocations, holdings, events	Baskets + isolated books + feed
Social	follows, comments, likes	Theme engagement

API Surface (16 Routers)

portfolio	analysis
risk-profile	stress-test
advisor	brief
stock-metrics	scores
themes	agent
paper	robinhood
news	market-macro
admin	community

Execution Modes

DRY RUN Intent logged, no order placed. Decision journal only.

PAPER Synthetic fills at market price. Same safety gates.

LIVE Real Robinhood orders via agentic MCP. Fenced account.

SUMMARY

Intelligence, not data.
Judgment, not charts.

9	30	44	9	~482
FEATURES LIVE	STRESS SCENARIOS	DATA MODELS	SAFETY GATES	TOKENS / TICK

Live at **emouva.com**

Built by Kondareddy Thanigundala

`React + TypeScript` `FastAPI` `Claude AI` `PostgreSQL` `Robinhood MCP`